Chapter 23

Campbell Biology, AP Edition, 12e

A finch.

Chapter 23

The Evolution of Populations

Figure 23.1

The medium ground finch (*Geospiza fortis*) is a seed-eating bird that inhabits the Galápagos Islands. In 1977, the *G. fortis* population on the island of Daphne Major was decimated by a long period of drought: Of some 1,200 birds, only 180 survived. The surviving birds had larger, deeper beaks, indicating that this population of finches had evolved.

What mechanisms can cause the evolution of populations?

Evolution occurs when natural selection, genetic drift, or gene flow alter the allele frequencies of a population over time. Genetic variation results in color variation in a hypothetical beetle population. Three mechanisms are given.

- Natural selection: The more visible darker beetles have been preyed on at a higher rate, causing lighter beetles (and the alleles they carry) to become more common over time.
- Genetic drift: By chance alone, alleles encoding darker color have become more common over several generations.
- Gene flow: The arrival of beetles with a spotted color pattern has brought new alleles into the population.

Key Concepts

- 23.1**Genetic variation makes evolution possible**
- 23.2**The Hardy-Weinberg equation can be used to test whether a population is evolving**
- 23.3**Natural selection, genetic drift, and gene flow can alter allele frequencies in a population**
- 23.4**Natural selection is the only mechanism that consistently causes adaptive evolution**

Advanced placement edition

Genetic variation within a population makes natural selection and evolution possible. Mathematics can be used to calculate changes **(BIG IDEA 1)** and the distribution of genes or alleles **(BIG IDEA 3)** within a population.

Study Tip

Make a table: This chapter discusses different types of selection—disruptive, stabilizing, sexual, balancing, frequency-dependent, and heterozygote advantage. These types of selection are not all mutually exclusive, and some evolutionary changes

are examples of more than one. To help you study, make a table like the one shown listing each type of selection, its distinguishing features, and example(s) from the chapter.

The table illustrates the type of selection, its distinguishing features, and example(s)

The table is as follows:

Type of selection	Distinguishing features	Example
Balancing selection	Maintains two or more phenotypes in a population, includes frequency-dependent selection and heterozygote advantage	Figures 23.17 and 23.18
Directional selection	blank	blank

A common misconception about evolution is that individual organisms evolve. It is true that natural selection acts on individuals: Each organism's traits affect its survival and reproductive success compared with other individuals. But the impact of natural selection only becomes apparent in how a *population* of organisms changes over time.

To see why, consider the population of medium ground finches decimated by a drought, resulting in only 180 survivors out of some 1,200 birds. Researchers Peter and Rosemary Grant observed that during the drought, small, soft seeds were scarce, while large, hard seeds were more plentiful. Birds with larger, deeper beaks were better able to crack and eat the larger seeds and survived at a higher rate than finches with smaller beaks.

Since beak depth is an inherited trait in these birds, the offspring of surviving birds also tended to have deep beaks. As a result, the average beak depth in the next generation of *G. fortis* was greater than it had been in the pre-drought population **(Figure 23.2)**. The finch population had evolved by natural selection. However, the *individual* finches did not evolve. Each bird had a beak of a particular size, which did not grow larger during the drought. Rather, the proportion of large beaks in the population increased from generation to generation: The population evolved, not its individual members.

Figure 23.2

Evidence of selection by food source.

Bar chart showing that the average beak depth (millimeters) for 1976 (similar to prior 3 years) is 8.9 and for 1978 (after drought) it is 9.7.

The data represent adult beak depths of finches hatched before and after the 1977 drought. In one generation, natural selection resulted in a larger average beak size in the population.

Instructors: A related Experimental Inquiry Tutorial can be assigned in **Mastering Biology**.

Focusing on evolutionary change in populations, we can define evolution on its smallest scale, called **microevolution**, as a change in allele frequencies in a population over generations. In the finch population, for example, birds that had alleles encoding large beaks survived at higher rates than did other birds—causing those alleles to be more common after the drought than they had been before it. There are three main mechanisms that can cause such changes in allele frequencies: natural selection, genetic drift (chance events that alter allele frequencies), and gene flow (the transfer of alleles between populations). Each of these mechanisms affects the genetic composition of populations, but only natural selection consistently improves the

degree to which organisms are well suited for life in their environment (adaptation). Before we examine this more closely, let's revisit a prerequisite for these processes: genetic variation.

Interview

Interview with Rosemary and Peter Grant: Studying evolution in the Galápagos finches**

A photograph of Rosemary and Perter Grant, evolutionary biologists studying Darwin's finches.

Get Ready for This Chapter

In this chapter, you will learn how natural selection, genetic drift, and gene flow can change the gene pool of a population, thus causing evolutionary change. Before beginning this chapter, you should be able to do the following:

- Distinguish between phenotype and genotype (see Concept 14.1, Figure 14.6).
- Use a Punnett square to determine the outcome of a genetic cross (see Figure 14.5).
- Evaluate the potential consequences of nucleotide-pair substitutions, insertions, and deletions (see Figure 17.26).

The following questions provide a quick check of your basic knowledge in these areas.

1. Suppose that a gene for coat color in mice has two alleles, B and b, where B is completely dominant and encodes a black coat color, and b is recessive and encodes a brown coat color. A mouse that is heterozygous has genotype _____, and its phenotype is _____.

BB; black

That's incorrect. Try again.

Incorrect. Review the following material: An organism's genotype refers to its genetic makeup, while its phenotype refers to its observable traits. In the case of a gene that has a completely dominant allele, organisms that are either homozygous for the dominant allele or heterozygous will display the dominant phenotype.

Bb; black

That's correct.

Correct!If you had trouble with this question, review the following material: An organism's genotype refers to its genetic makeup, while its phenotype refers to its observable traits. A mouse that is heterozygous has two different alleles and thus has a genotype of Bb. Since the B allele is completely dominant and encodes a black coat color, the mouse has a black (coat color) phenotype.

bb; brown

That's incorrect. Try again.

Incorrect. Review the following material: An organism's genotype refers to its genetic makeup, while its phenotype refers to its observable traits. In the case of a gene that has a completely dominant allele, organisms that are either homozygous for the dominant allele or heterozygous will display the dominant phenotype.

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2. Suppose a BB female mouse mates with a Bb male mouse. Which of the following represents the probabilities of each genotype occurring among their offspring?

one quarter BB, one half Bb, one quarter bb

That's incorrect. Try again.

Incorrect. Review the following material: Each egg of a BB female mouse will carry the B allele, so all the offspring will have at least one B allele. Half of the sperm of a Bb male mouse are predicted to carry the B allele, while the other half are predicted to carry the b allele.

one half BB, one half Bb

That's correct.

Correct! If you had trouble with this question, review the following material: Each egg of a BB female mouse will carry the B allele, so all the offspring will have at least one B allele. Half of the sperm of a Bb male mouse are predicted to carry the B allele, while the other half are predicted to carry the b allele. Therefore, assuming gametes combine randomly, half of the offspring would be BB and half of the offspring would be Bb.

one half Bb, one half bb

That's incorrect. Try again.

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3. Consider the following mutations to the nucleotide sequence of a gene. Which of these would be a frameshift mutation?

a deletion that removes 3 nucleotide pairs

That's incorrect. Try again.

Incorrect. Review the following material: Nucleotide-pair insertions or deletions that alter the reading frame of a gene's genetic message (the triplet grouping of nucleotides on the mRNA that is read during translation) are called frameshift mutations. A frameshift mutation occurs whenever the number of nucleotide pairs that is inserted or deleted is not a multiple

of three. When a frameshift mutation occurs, all of the amino acids encoded by the codons after the mutation will be altered, and eventually a stop codon will occur. Thus, the protein produced will be abnormal and likely nonfunctional. In contrast, if the number of nucleotide pairs that is inserted or deleted equals three (or is a multiple of three), one (or more) amino acids will be inserted or deleted; this can cause a variety of effects, but it does not alter the reading frame of the gene's genetic message. Similarly, a single nucleotide-pair substitution—a mutation in which one nucleotide and its partner are replaced by another pair of nucleotides—can have a variety of effects, but it does not alter the triplet grouping of nucleotides to be read during translation.

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a single nucleotide-pair insertion
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That's correct.

Correct! If you had trouble with this question, review the following material: Nucleotide-pair insertions or deletions that alter the reading frame of a gene's genetic message (the triplet grouping of nucleotides on the mRNA that is read during translation) are called frameshift mutations. A frameshift mutation occurs whenever the number of nucleotide pairs that is inserted or deleted is not a multiple of three. When a frameshift mutation occurs, all of the amino acids encoded by the codons after the mutation will be altered, and eventually a stop codon will occur. Thus, the protein produced will be abnormal and likely nonfunctional. In contrast, if the number of nucleotide pairs that is inserted or deleted equals three (or is a multiple of three), one (or more) amino acids will be inserted or deleted; this can cause a variety of effects, but it does not alter the reading frame of the gene's genetic message. Similarly, a single nucleotide-pair substitution—a mutation in which one nucleotide and its partner are replaced by another pair of nucleotides—can have a variety of effects, but it does not alter the triplet grouping of nucleotides to be read during translation. Note that the insertion or deletion of one or more amino acids (which occurs when a multiple of three nucleotide pairs are inserted or deleted) can have harmful effects. Likewise, a single nucleotide-pair substitution can have harmful effects if it produces a new stop codon, or if it results in the replacement of a key amino acid with a different amino acid that causes the resulting protein to be abnormal (and likely nonfunctional). Other single nucleotide-pair substitutions may have little or no effect if they occur in a noncoding region of the gene or if they result in a silent mutation (one which does not alter the amino acid sequence).

a single nucleotide-pair substitution in which an A-T nucleotide pair is replaced by a G-C nucleotide pair

That's incorrect. Try again.

Incorrect. Review the following material: Nucleotide-pair insertions or deletions that alter the reading frame of a gene's genetic message (the triplet grouping of nucleotides on the mRNA that is read during translation) are called frameshift mutations. A frameshift mutation occurs whenever the number of nucleotide pairs that is inserted or deleted is not a multiple of three. When a frameshift mutation occurs, all of the amino acids encoded by the codons after the mutation will be altered, and eventually a stop codon will occur. Thus, the protein produced will be abnormal and likely nonfunctional. In contrast, if the number of nucleotide pairs that is inserted or deleted equals three (or is a multiple of three), one (or more) amino acids will be inserted or deleted; this can cause a variety of effects, but it does not alter the reading frame of the gene's genetic message. Similarly, a single nucleotide-pair substitution—a mutation in which one nucleotide and its partner are replaced by another pair of nucleotides—can have a variety of effects, but it does not alter the triplet grouping of nucleotides to be read during translation.

a single nucleotide-pair substitution in which an A-T nucleotide pair is replaced by a T-A nucleotide pair

That's incorrect. Try again.

Incorrect. Review the following material: Nucleotide-pair insertions or deletions that alter the reading frame of a gene's genetic message (the triplet grouping of nucleotides on the mRNA that is read during translation) are called frameshift mutations. A frameshift mutation occurs whenever the number of nucleotide pairs that is inserted or deleted is not a multiple of three. When a frameshift mutation occurs, all of the amino acids encoded by the codons after the mutation will be altered, and eventually a stop codon will occur. Thus, the protein produced will be abnormal and likely nonfunctional. In contrast, if the number of nucleotide pairs that is inserted or deleted equals three (or is a multiple of three), one (or more) amino acids will be inserted or deleted; this can cause a variety of effects, but it does not alter the reading frame of the gene's genetic message. Similarly, a single nucleotide-pair substitution—a mutation in which one nucleotide and its partner are replaced by another pair of nucleotides—can have a variety of effects, but it does not alter the triplet grouping of nucleotides to be read during translation.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 23.1Describe how genetic variation arises and explain why it is a prerequisite for evolution.
- 23.2Use allele-frequency data to predict the genotype frequencies of a population in Hardy-Weinberg equilibrium.
- 23.3Differentiate between how natural selection, genetic drift, and gene flow affect allele frequencies in a population.
- 23.4Explain how natural selection can lead to adaptation.

Concept 23.1: Genetic variation makes evolution possible

In *The Origin of Species*, Darwin provided abundant evidence that life on Earth has evolved over time, and he proposed natural selection as the primary mechanism for that change. He observed that individuals differ in their inherited traits and that selection acts on such differences, leading to evolutionary change. Although Darwin realized that variation in heritable traits is a prerequisite for evolution, he did not know precisely how organisms pass heritable traits to their offspring.

Just a few years after Darwin published *The Origin of Species*, Gregor Mendel wrote a groundbreaking paper on inheritance in pea plants proposing a model of inheritance in which organisms transmit discrete heritable units (now called genes) to their offspring. Although Darwin did not know about genes, Mendel's paper set the stage for understanding the genetic differences on which evolution is based.

Genetic Variation

Individuals within all species vary in their phenotypic traits. Humans, for example, vary noticeably in facial features, height, and voice. And though you cannot identify blood group (A, B, AB, or O) from a person's appearance, this and many other molecular traits also vary extensively among individuals.

Such phenotypic variations often reflect **genetic variation**, differences among individuals in the composition of their genes or other DNA sequences. Some heritable phenotypic differences occur on an "either-or" basis, such as the flower colors of Mendel's pea plants: Each plant had flowers that were either purple or white (see Figure 14.3). Characters that vary in this way are typically determined by a single gene locus, with different alleles producing distinct phenotypes. In contrast, other phenotypic differences vary in gradations along a continuum. Such variation usually results from the influence of two or more

genes on a single phenotypic character. In fact, many phenotypic characters are influenced by multiple genes, including coat color in horses **(Figure 23.3)**, seed number in maize (corn), and height in humans.

Figure 23.3

Phenotypic variation in horses.

Horses with several coat colors are shown: white, red, brown, dark brown.

How much do genes and other DNA sequences vary from one individual to another? Genetic variation at the whole-gene level (*gene variability*) can be quantified as the average percentage of loci that are heterozygous. (Recall that a heterozygous individual has two different alleles for a given locus, whereas a homozygous individual has two identical alleles for that locus.) As an example, on average the fruit fly *Drosophila melanogaster* is heterozygous for about 1,920 of its 13,700 loci 14% and homozygous for all the rest.

Considerable genetic variation can also be measured at the molecular level of DNA (*nucleotide variability*). But little of this variation results in phenotypic variation. Why? Many nucleotide variations occur within *introns*, noncoding segments of DNA lying between *exons*, the regions retained in mRNA after RNA processing (see Figure 17.12). And of the variations that occur within exons, most do not cause a change in the amino acid sequence of the protein encoded by the gene. For example, in the sequence comparison shown in **Figure 23.4**, there are 43 nucleotide sites with variable base pairs (where substitutions have occurred), as well as several sites where insertions or deletions have occurred. Although 18 variable sites occur within the four exons of the *Adh* gene, only one of these variations (at site 1,490) results in an amino acid change. Note, however, that this single variable site is enough to cause genetic variation at the level of the gene, and hence two different forms of the Adh enzyme are produced.

Figure 23.4

Extensive genetic variation at the molecular level.

D N A sequence variations in alcohol dehydrogenase gene of Drosophila melanogaster.

An image of the comparative base sequence of the alcohol dehydrogenase gene of fruit flies. Both exons (coding regions which are translated into amino acids) and introns (non-coding region between exons) are depicted. Base substitutions, insertions, and deletions are represented.

The scale from left to right is from 1 to 2,500 base pairs. Exon regions are depicted at (about) 1 to 100; 700 to 900; 950 to 1,300; and 1,400 to 1,900, base pairs, respectively. Substitution at 1,500 base pairs results in the translation of a different amino acid. Five insertion sites are shown at 400, 550, 1750, 2100, and 2,300 base pairs. A deletion of 26 base pairs occurs between 2,400 and 2,450 base pairs. Sequence comparison shows that there are 43 nucleotide sites with variable base pairs (where substitutions have occurred) as well as five sites where insertions or deletions have occurred.

This diagram summarizes data from a study comparing the DNA sequence of the alcohol dehydrogenase (*Adh*) gene in several fruit flies (*Drosophila melanogaster*). The *Adh* gene has four exons (dark blue) separated by introns (light blue); the exons include the coding regions that are ultimately translated into the amino acids of the Adh enzyme (see Figure 5.1). Only one substitution has a phenotypic effect, producing a different form of the Adh enzyme.

MAKE CONNECTIONS*Review**Figures 17.6**and*17.11*. Explain how a base-pair substitution that alters a coding region of the* Adh *locus could have no effect on amino acid sequence. Then explain how an insertion in an exon could have no effect on the protein produced.*

*For suggested answer, see*Appendix A.**

It is important to bear in mind that some phenotypic variation does not result from genetic differences among individuals (**Figure 23.5** shows a striking example in a caterpillar of the southwestern United States). Phenotype is the product of an inherited genotype and many environmental influences (see Concept 14.3). In a human example, bodybuilders alter their phenotypes dramatically but do not pass their huge muscles on to the next generation. In general, only the genetically determined part of phenotypic variation can have evolutionary consequences. As such, genetic variation provides the raw material for evolutionary change: *Without genetic variation, evolution cannot occur*.

Figure 23.5

Nonheritable variation.

Photos of two types of caterpillars. a. Caterpillars resembling oak flowers b. Caterpillars resembling oak twigs.

These caterpillars of the moth Nemoria arizonaria owe their different appearances to chemicals in their diets, not to differences in their genotypes. ***(a)*** *Caterpillars raised on a diet of oak flowers resemble the flowers, whereas* ***(b)*** *their siblings raised on oak leaves resemble oak twigs.*

Sources of Genetic Variation

Genetic variation originates when mutation, gene duplication, or other processes produce new alleles and new genes. Genetic variants can be produced rapidly in organisms with short generation times. Sexual reproduction can also result in genetic variation as existing genes are arranged in new ways.

Formation of New Alleles

New alleles can arise by *mutation*, a change in the nucleotide sequence of an organism's DNA. Mutations can be caused by factors such as errors in DNA replication, exposure to UV light and other high-energy forms of radiation, and exposure to certain chemicals (see Concept 17.5). A change of as little as one base in a gene—a "point mutation"—can have a significant impact on phenotype, as in sickle-cell disease (see Figure 23.18). Furthermore, organisms reflect many generations of past selection, and hence their phenotypes tend to be well suited for life in their environments. As a result, most new mutations that alter a phenotype are at least slightly harmful.

In some cases, natural selection quickly removes such harmful alleles. In diploid organisms, however, harmful alleles that are recessive can be hidden from selection. Indeed, a harmful recessive allele can persist for generations by propagation in heterozygous individuals (where its harmful effects can be masked by the more favorable dominant allele). Such "heterozygote protection" maintains a huge pool of alleles that may be harmful under present conditions, but that could be beneficial if the environment changes.

While many mutations are harmful, many others are not. Recall that much of the DNA in eukaryotic genomes does not encode proteins (see Figure 21.6). Point mutations in these noncoding regions generally result in **neutral variation**, differences in DNA sequence that do not confer a selective advantage or disadvantage. The redundancy in the genetic code is another source of neutral variation: Even a point mutation in a gene that encodes a protein will have no effect on the protein's function if the amino acid doesn't change. And even a change in the amino acid may not affect the protein's shape and function. Moreover, a mutant allele may on rare occasions actually make its bearer better suited to the environment, enhancing reproductive success.

Finally, note that in multicellular organisms, only mutations in cell lines that produce gametes can be passed to offspring. In plants and fungi, this is not as limiting as it may sound, since many different cell lines can produce gametes. But in most animals, the majority of mutations occur in somatic cells and are not passed to offspring.

Altering Gene Number or Position

Chromosomal changes that delete, disrupt, or rearrange many loci are usually harmful. However, when such large-scale changes leave genes intact, they may not affect the organisms' phenotype. In rare cases, chromosomal rearrangements may even be beneficial. For example, the translocation of part of one chromosome to a different chromosome could link genes in a way that produces a positive effect.

A key potential source of variation is the duplication of genes due to errors in meiosis (such as unequal crossing over), slippage during DNA replication, or the activities of transposable elements (see Concept 21.5). Duplications of large chromosome segments, like other chromosomal aberrations, are often harmful, but the duplication of smaller pieces of DNA may not be. Gene duplications that do not have severe effects can persist over generations, allowing mutations to accumulate. The result is an expanded genome with new genes that may take on new functions.

Such increases in gene number appear to have played a major role in evolution. For example, the remote ancestors of mammals had a single gene for detecting odors that has since been duplicated many times. As a result, humans today have about 380 functional olfactory receptor genes, and mice have about 1,200. This dramatic proliferation of olfactory genes probably helped early mammals, enabling them to detect faint odors and to distinguish among many different smells.

Rapid Reproduction

Mutation rates tend to be low in plants and animals, averaging about one mutation in every 100,000 genes per generation, and are often even lower in prokaryotes. But prokaryotes have many more generations per unit of time, so mutations can quickly generate genetic variation in their populations. The same is true of viruses. For instance, HIV has a generation time of about two days (that is, it takes two days for a newly formed virus to produce the next generation). HIV also has an RNA genome, which has a much higher mutation rate than a typical DNA genome due to the lack of RNA repair mechanisms in host cells (see Concept 19.2). Thus, single-drug treatments are less effective against HIV: Mutant forms of the virus, resistant to a particular drug, tend to proliferate quickly. The most effective AIDS treatments to date have been drug "cocktails" that combine several medications. This approach has worked well as it is less likely that a set of mutations that together confer resistance to *all* the drugs will occur in a short time period.

Sexual Reproduction

In organisms that reproduce sexually, most of the genetic variation in a population results from the unique combination of alleles that each individual receives from its parents. Of course, at the nucleotide level, all the differences among these alleles have originated from past mutations. Sexual reproduction then shuffles existing alleles and deals them at random to produce individual genotypes.

Three mechanisms contribute to this shuffling: crossing over, independent assortment of chromosomes, and fertilization (see Concept 13.4). During meiosis, homologous chromosomes, one inherited from each parent, trade some of their alleles by crossing over. The resulting recombinant chromosomes are then distributed at random into gametes. Then, because myriad possible mating combinations exist in a population, fertilization typically brings together gametes with different genetic backgrounds. Together, these three mechanisms ensure that sexual reproduction rearranges existing alleles into fresh combinations each generation, providing much of the genetic variation that makes evolution possible.

Animation

Origins of Genetic Variation

image

Play

image

Play

Concept Check 23.1

- Explain why genetic variation within a population is a prerequisite for evolution.
- Of all the mutations that occur in a population, why do only a small fraction become widespread?
- **MAKE CONNECTIONS** If a population stopped reproducing sexually but still reproduced asexually, how would its genetic variation be affected over time? (See Concept 13.4.)

*For suggested answers, see*Appendix A.**

Concept 23.2: The Hardy-Weinberg equation can be used to test whether a population is evolving

The presence of genetic variation in a population does not guarantee that the population will evolve. For evolution to occur, one or more factors must be at work. In this section, we'll explore one way to test whether evolution is occurring in a population.

Gene Pools and Allele Frequencies

A **population** is a group of individuals of the same species that live in the same area and interbreed, producing fertile offspring. Different populations of a species may be isolated geographically from one another, exchanging genetic material only rarely. This is common for species that live on widely separated islands or in different lakes, but not all populations are isolated **(Figure 23.6)**. Still, members of a population typically breed with one another and thus are more closely related to each other than to members of other populations.

Figure 23.6

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One species, two populations.
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Porcupine herd of caribou ranges from northern Alaska and Canada bordering the Arctic Ocean; Forty mile herd range shares the southern reaches of the porcupine herd but travels further south.

These two caribou populations in the Yukon are not totally isolated; they sometimes share the same area (shaded purple). Still, members of each population are most likely to breed within their own population.

Porcupine herd of caribou range from northern Alaska and Canada bordering the Arctic Ocean; Forty mile herd range shares the southern reaches of the porcupine herd but travels further south. The porcupine herd is larger caribou with antlers, the Forty mile herd is smaller without antlers.

We can characterize a population's genetic makeup by describing its **gene pool**, which consists of all copies of every type of allele at every locus in all members of the population. If only one allele exists for a particular locus in a population, that allele is said to be *fixed* in the gene pool, and all individuals are homozygous for that allele. But if there are two or more alleles for a particular locus in a population, individuals may be either homozygous or heterozygous.

For example, imagine a population of 500 wildflower plants with two alleles, CR and CW, for a locus that codes for flower pigment. These alleles show incomplete dominance; thus, each genotype has a distinct phenotype. Plants homozygous for the CR allele (CRCR) produce red pigment and have red flowers; plants homozygous for the CW allele (CWCW) produce no red pigment and have white flowers; and heterozygotes (CRCW) produce some red pigment and have pink flowers.

A red flower is C superscript R, C superscript R; a white flower is C superscript W, C superscript W; a pink flower is C superscript R, C superscript W.

Each genotype and each allele has a frequency (proportion) in the population. To see how these frequencies are calculated, suppose our population has 320 plants with red flowers (genotype CRCR), 160 with pink flowers (genotype CRCW), and 20 with white flowers (genotype CWCW). Since 320 of the 500 plants have red flowers, the frequency of the CRCR genotype is 0.64 (320/500). Likewise, the frequency of the CRCW genotype is 0.32 (160/500) and the frequency of the CWCW genotype is 0.04 (20/500).

With respect to allele frequencies, wildflowers are diploid organisms, so each individual has two alleles for each of its genes. Hence, the 500 individuals in our population have a total of 1,000 copies of the gene for flower color. The CR allele accounts for 800 of these copies (320×2=640 for CRCR plants, plus 160×1=160 for CRCW plants). Thus, the frequency of the CR allele is 800/1,000=0.8 (80%).

When studying a locus with two alleles, the convention is to use p to represent the frequency of one allele and q to represent the frequency of the other allele. Thus, p, the frequency of the CR allele in the gene pool of this population, is p=0.8(80%). And because there are only two alleles for this gene, the frequency of the CW allele, represented by q, must be q=1−p=0.2(20%). For loci that have more than two alleles, the sum of all allele frequencies must still equal 1100%.

The Hardy-Weinberg Equation

One way to assess whether natural selection or other factors are causing evolution at a particular locus is to determine what the genetic makeup of a population would be if it were *not* evolving at that locus. We can then compare that with the data we actually observed for the population. If there are no differences, this suggests that the population is not evolving. If there are differences, this suggests that the population is evolving—and then we can investigate why.

Hardy-Weinberg Equilibrium

In a population that is not evolving, allele and genotype frequencies will remain constant from generation to generation, provided that only Mendelian segregation and recombination of alleles are at work. Such a population is said to be in **Hardy-**

Weinberg equilibrium, named for the British mathematician and German physician, respectively, who independently developed this idea in 1908.

To determine whether a population is in Hardy-Weinberg equilibrium, it is helpful to think about genetic crosses in a new way. Previously, we used Punnett squares to determine the genotypes of offspring in a genetic cross (see Figure 14.5). Here, instead of considering the possible allele combinations from one cross, we'll consider the combination of alleles in *all* of the crosses in a population. Imagine that all the alleles for a given locus from all the individuals in a population are placed in a large bin **(Figure 23.7)**. We can think of this bin as holding the population's gene pool for that locus. "Reproduction" occurs by selecting alleles at random from the bin; somewhat similar events occur in nature when fishes release sperm and eggs into the water or when plant sperm (in pollen) is blown about by the wind. By viewing reproduction as a process of randomly selecting and combining alleles from the bin (the gene pool), we are in effect assuming that mating occurs at random: that all male-female matings are equally likely.

Figure 23.7

Selecting alleles at random from a gene pool.

An illustration of the Hardy-Weinberg equilibrium of allele frequencies in a population.

Random selection of alleles from a gene pool following Hardy-Weinberg equilibrium shows that the gametes produced also follow the same ratios for allele frequency. The illustration is divided as follows:

- Frequencies of alleles:

The allele frequencies of the population are 0.8 (80 percent) and 0.2 (20 percent). p equals frequency of C superscript R allele (red) equals 0.8.

q equals frequency of C superscript W allele (white) equals 0.2.

- Alleles in the population:

If all the alleles could be placed in a large bin (representing the gene pool), 80 percent would be C superscript R and 20 percent would be C superscript W.

- Gametes produced:

Assuming mating is random, each time two gametes come together, there is an 80 percent chance the egg carries a C superscript R allele and a 20 percent chance it carries a C superscript W allele.

- Likewise, each sperm has an 80 percent chance of carrying a C superscript R allele and a 20 percent chance of carrying a C superscript W allele.

DRAW IT*Draw a bin with six white balls instead of four. For the frequency of*CR*to remain equal to*0.8*, how many red balls should the bin contain?*

*For suggested answer, see**Appendix A.**

BioFlix® Animation

Allele Frequencies

image

image

Let's apply the bin analogy to the hypothetical wildflower population discussed earlier. In that population of 500 flowers, the frequency of the allele for red flowers (CR) is p=0.8, and the frequency of the allele for white flowers (CW) is q=0.2. This implies that a bin holding all 1,000 copies of the flower-color gene in the population would contain 800CR alleles and 200CW alleles. Assuming that gametes are formed by selecting alleles at random from the bin, the probability that an egg or sperm contains a CR or CW allele is equal to the frequency of these alleles in the bin. Thus, as shown in Figure 23.7, each egg (and each sperm) has an 80% chance of containing a CR allele and a 20% chance of containing a CW allele.

Using the rule of multiplication (see Figure 14.9), we can now calculate the frequencies of the three possible genotypes, assuming random unions of sperm and eggs. The probability that two CR alleles will come together is p×p=p2=0.8×0.8=0.64. Thus, about 64% of the plants in the next generation will have the genotype CRCR. The frequency of CWCW individuals is expected to be about q×q=q2=0.2×0.2=0.04, or 4%. CRCW heterozygotes can arise in two different ways. If the sperm provides the CR allele and the egg provides the CW allele, the resulting heterozygotes will be p×q=0.8×0.2=0.16, or 16% of the total. If the sperm provides the CW allele and the egg the CR allele, the heterozygous offspring will make up q×p=0.2×0.8=0.16, or 16%. The frequency of heterozygotes is thus the sum of these possibilities: pq+qp=2pq=0.16+0.16=0.32, or 32%.

As shown in **Figure 23.8**, the genotype frequencies in the next generation must add up to 1(100%). Thus, the equation for Hardy-Weinberg equilibrium states that if a population is not evolving at a locus with two alleles, the three genotypes will appear in the following proportions:

p2+2pq+q2=1ExpectedExpectedExpectedfrequencyfrequencyfrequencyof genotypeof genotypeof genotypeCRCRCRCWCWCW

Figure 23.8

Hardy-Weinberg equilibrium.

Illustration of Hardy-Weinberg equilibrium through generations of wildflowers.

In our wildflower population, the gene pool remains constant from one generation to the next. Mendelian processes alone do not alter frequencies of alleles or genotypes.

The gene pool of the initial (parental) generation consists of: 80% C superscript R (p = 0.8) crossed with 20% C superscript W (q = 0.2). A Punnett square is as follows:

- First row: C superscript R (p = 0.8) egg crossed with C superscript R (p= 0.8 sperm) = 0.64 (p squared) C superscript R, C superscript R. Phenotype: Red.

C superscript R (p = 0.8) egg crossed with C superscript W sperm (p = 0.2) = 0.16 (pq) C superscript R, C superscript W. Phenotype: pink.

- Second row: C superscript W (q = 0.2) egg crossed with C superscript R (p= 0.8 sperm) = 0.16 (q p) C superscript R, C superscript W. Phenotype = pink.

C superscript W (q = 0.2) egg crossed with C superscript W sperm (p = 0.2) = 0.04 (q squared) C superscript W, C superscript W. Phenotype = white.

- Genotypes in this offspring generation are: 64% C superscript R, C superscript R; 32% C superscript R, C superscript W and 4% C superscript W, C superscript W.

- Gametes in this offspring generation go to: 64% C superscript R (from C superscript R, C superscript R plants) + 16% C superscript R (from C superscript R, C superscript W plants) = 80% C superscript R = 0.8 = p.

4% C superscript W (from C superscript W, C superscript W plants) + 16% C superscript W (from C superscript R, C superscript W plants) = 20% C superscript W = 0.2 = q.

- With random mating, these gametes will result in the same mix of genotypes in the next generation: 64% C superscript R, C superscript R; 32% C superscript R, C superscript W and 4% C superscript R, C superscript W.

WHAT IF?If the frequency of theCR*allele were0.6*, predict the frequencies of the*CRCR, CRCW, andCWCW*genotypes.*

*For suggested answer, see**Appendix A*.*

Figure 23.8 Walkthrough

Hardy-Weinberg equilibrium

Watch an explanation of this figure in the Figure 23.8 Walkthrough: Hardy-Weinberg equilibrium.

image

Note that for a locus with two alleles, only three genotypes are possible (in this case, CRCR, CRCW, and CWCW). As a result, the sum of the frequencies of the three genotypes must equal 1(100%) in *any* population regardless of whether the population is in Hardy-Weinberg equilibrium. The key point is that a population is in Hardy-Weinberg equilibrium only if the observed genotype frequency of one homozygote is p2, the observed frequency of the other homozygote is q2, and the observed frequency of heterozygotes is 2pq. Finally, as suggested by Figure 23.8, if a population such as our wildflowers is in Hardy-Weinberg equilibrium and its members continue to mate randomly generation after generation, allele and genotype frequencies will remain constant. The system operates somewhat like a deck of cards: No matter how many times the deck is reshuffled to deal out new hands, the deck itself remains the same. Aces do not grow more numerous than jacks. And the repeated shuffling of a population's gene pool over the generations cannot, in itself, change the frequency of one allele relative to another.

Conditions for Hardy-Weinberg Equilibrium

The Hardy-Weinberg approach describes a population that is not evolving. This can occur if a population meets all five of the conditions for Hardy-Weinberg equilibrium listed in **Table 23.1**. But in nature, the allele and genotype frequencies of a population often *do* change over time. Such changes can occur when at least one of the conditions for Hardy-Weinberg equilibrium is not met.

Table 23.1

Conditions for Hardy-Weinberg Equilibrium

Table 23.1, Conditions for Hardy-Weinberg Equilibrium .

The table is as follows:

Condition	Consequence if condition does not hold
1. No mutation	The gene pool is modified if mutations occur or if entire genes are deleted or duplicated.

2. Random mating	If individuals mate within a subset of the population, such as near neighbors or close relatives (inbreeding), random mixing of gametes does not occur and genotype frequencies change.
3. No natural selection	Allele frequencies change when individuals with different genotypes show consistent differences in their survival or reproductive success.
4. Extremely large population size	In small populations, allele frequencies fluctuate by change over time (genetic drift)
5. No gene flow	By moving alleles into or out of populations, gene flow can alter allele frequencies

Animation

Causes of Evolutionary Change

**

Although departure from the conditions in Table 23.1 is common—resulting in evolutionary change—it is also common for natural populations to be in Hardy-Weinberg equilibrium for specific genes. One way this can happen is if selection alters allele frequencies at some loci but not others. In addition, some populations evolve so slowly that the changes in their allele and genotype frequencies are difficult to distinguish from those predicted for a non-evolving population.

CheckPoint

Using the Hardy-Weinberg Equation

Check your understanding before you move on, with the CheckPoint: Using the Hardy-Weinberg Equation.

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Applying the Hardy-Weinberg Equation

The Hardy-Weinberg equation is often used as an initial test of whether evolution is occurring in a population (Concept Check 23.2, question 3 is an example). The equation also has medical applications, such as estimating the percentage of a population carrying the allele for an inherited disease. For example, consider phenylketonuria (PKU), a metabolic disorder that results from homozygosity for a recessive allele. This disorder occurs in about one out of every 10,000 babies born in the United States. Left untreated, PKU results in mental disability and other problems. (As described in Concept 14.4, newborns are now routinely tested for PKU, and symptoms can be largely avoided with a diet very low in phenylalanine.)

To apply the Hardy-Weinberg equation, we must assume that no new PKU mutations are being introduced into the population (condition 1) and that people neither choose their mates on the basis of whether or not they carry this gene nor generally mate with close relatives (condition 2). We must also ignore any effects of differential survival and reproductive success among PKU genotypes (condition 3) and assume that there are no effects of genetic drift (condition 4) or of gene flow from other populations into the United States (condition 5). These assumptions are reasonable: The mutation rate for the PKU gene is low, inbreeding and other forms of nonrandom mating are not common in the United States, selection occurs only against the rare homozygotes (and then only if dietary restrictions are not followed), the U.S. population is very large, and populations outside the country have PKU allele frequencies similar to those seen in the United States.

If all these assumptions hold, then the frequency of individuals in the population born with PKU will correspond to q2 in the Hardy-Weinberg equation (q2=frequency of homozygotes). Because the allele is recessive, we must estimate the number of heterozygotes rather than counting them directly as we did with the pink flowers. Recall that there is one PKU occurrence per 10,000 births, which indicates that q2=0.0001. Thus, the frequency (q) of the recessive allele for PKU is

q=0.0001=0.01

and the frequency of the dominant allele is

p=1−q=1−0.01=0.99

The frequency of carriers, heterozygous people who do not have PKU but may pass the PKU allele to offspring, is

2pq=2×0.99×0.01=0.0198(approximately 2% of the U.S. population)

Remember, the assumption of Hardy-Weinberg equilibrium yields an approximation; the real number of carriers may differ. Still, our calculations suggest that harmful recessive alleles at this and other loci can be concealed in a population because they are carried by healthy heterozygotes. The **Scientific Skills Exercise** provides another opportunity for you to apply the Hardy-Weinberg equation to allele data.

Scientific Skills Exercise

Using the Hardy-Weinberg Equation to Interpret Data and Make Predictions

AP Science Practices 2, 3, 4, 5, 6

Is Evolution Occurring in a Soybean Population? One way to test whether evolution is occurring in a population is to compare the observed genotype frequencies at a locus with those expected for a non-evolving population based on the Hardy-Weinberg equation. In this exercise, you will test whether a soybean population is evolving at a locus with two alleles, CG and CY, that affect chlorophyll production and hence leaf color.

Green soybeans.

How the Experiment Was Done Students planted soybean seeds and then counted the number of seedlings of each genotype at day 7 and again at day 21. Seedlings of each genotype could be distinguished visually because the CG and CY alleles show incomplete dominance: CGCG seedlings have green leaves, CGCY seedlings have green-yellow leaves, and CYCY seedlings have yellow leaves.

Data from the Experiment

A table of experiment results.

Time (days)	Number of Seedlings
Green (C superscript G C superscript G)	Green-yellow (C superscript G C superscript Y)

7	49
21	47

INTERPRET THE DATA

- Use the observed genotype frequencies from the day 7 data to calculate the frequencies of the CG allele (p) and the CY allele (q).
- Next, use the Hardy-Weinberg equation (p2+2pq+q2=1) to calculate the day 7 expected frequencies of genotypes CGCG, CGCY, and CYCY for a population in Hardy-Weinberg equilibrium.
-
  - Calculate the observed frequencies of genotypes CGCG, CGCY, and CYCY at day 7.
- Compare these frequencies to the expected frequencies calculated in question 2. Is the seedling population in Hardy-Weinberg equilibrium at day 7, or is evolution occurring? Explain your reasoning and identify which genotypes, if any, appear to be selected for or against.
-
  - Calculate the observed frequencies of genotypes CGCG, CGCY, and CYCY at day 21.
- Compare these frequencies to the expected frequencies calculated in question 2 and to the observed frequencies at day 7. Is the seedling population in Hardy-Weinberg equilibrium at day 21, or is evolution occurring? Explain your reasoning and identify which genotypes, if any, appear to be selected for or against.
- Homozygous CYCY individuals cannot produce chlorophyll. The ability to photosynthesize becomes more critical as seedlings age and begin to exhaust the supply of food that was stored in the seed from which they emerged. Develop a hypothesis that explains the data for days 7 and 21. Based on this hypothesis, predict how the frequencies of the CG and CY alleles will change beyond day 21.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Concept Check 23.2

- A population has 85 individuals of genotype AA, 320 of genotype Aa, and 295 of genotype aa. Calculate the genotype frequencies and the frequencies of alleles A and a.
- The frequency of allele a is 0.45 for a population in Hardy-Weinberg equilibrium. What are the expected frequencies of genotypes AA, Aa, and aa?
- **WHAT IF?** A locus that affects susceptibility to a degenerative brain disease has two alleles, V and v. In a population, 16 people have genotype VV, 92 have genotype Vv, and 12 have genotype vv. Is this population evolving? Explain.

*For suggested answers, see*Appendix A.**

Concept 23.3: Natural selection, genetic drift, and gene flow can alter allele frequencies in a population

Note again the five conditions required for a population to be in Hardy-Weinberg equilibrium (see Table 23.1). A deviation from any of these conditions is a potential cause of evolution. New mutations (violation of condition 1) can alter allele frequencies, but because mutations are rare, the change from one generation to the next is likely to be very small. Nonrandom mating (violation of condition 2) can affect the frequencies of homozygous and heterozygous genotypes but by itself has no

effect on allele frequencies in the gene pool. (Allele frequencies can change if individuals with certain inherited traits are more likely than other individuals to obtain mates. However, such a situation not only causes a deviation from random mating, but also violates condition 3, no natural selection.)

For the rest of this section we will focus on the three mechanisms that alter allele frequencies directly and cause most evolutionary change: natural selection, genetic drift, and gene flow (violations of conditions 3–5).

Natural Selection

The concept of natural selection is based on differential success in survival and reproduction: Individuals in a population exhibit variations in their heritable traits, and those with traits that are better suited to their environment tend to produce more offspring than those with traits that are not as well suited.

In genetic terms, selection results in alleles being passed to the next generation in proportions that differ from the present generation. For example, the fruit fly *D. melanogaster* has an allele that confers resistance to the insecticide DDT. This allele has a frequency of 0% in laboratory strains of *D. melanogaster* from flies collected in the wild in the early 1930s, prior to DDT use. However, in strains established from flies collected after 1960 (following 20 or more years of DDT use), the allele frequency is 37%. Thus, this allele either arose by mutation between 1930 and 1960 or was present in 1930, but very rare. In any case, the rise in frequency of this allele most likely occurred because DDT is an effective poison that is a strong selective force in exposed fly populations.

As the *D. melanogaster* example suggests, an allele that confers resistance to an insecticide will increase in frequency in a population exposed to that insecticide. Such changes are not coincidental. By consistently favoring some alleles over others, natural selection can cause **adaptive evolution**, in which traits that enhance survival or reproduction tend to increase in frequency over time. We'll explore this process in Concept 23.4.

Genetic Drift

If you flip a coin 1,000 times, a result of 700 heads and 300 tails might make you suspicious about that coin. But if you flip a coin only 10 times, an outcome of 7 heads and 3 tails would not be surprising. The smaller the number of coin flips, the more likely it is that chance alone will cause a deviation from the predicted result. (In this case, the prediction is an equal number of heads and tails.) Chance events can also cause allele frequencies to fluctuate unpredictably from one generation to the next, especially in small populations—a process called **genetic drift**.

Figure 23.9 models how genetic drift might affect a small population of our wildflowers. In this example, drift leads to the loss of an allele from the gene pool, but it is a matter of chance that the CW allele is lost and not the CR allele. Such unpredictable changes in allele frequencies can be caused by chance events associated with survival and reproduction. Perhaps a large animal such as a moose stepped on the three CWCW individuals in generation 2, killing them and increasing the chance that only the CR allele would be passed to the next generation. Allele frequencies can also be affected by chance events that occur during fertilization. For example, suppose two individuals of genotype CRCW had a small number of offspring. By chance alone, every egg and sperm pair that generated offspring could happen to have carried the CR allele and not the CW allele.

Figure 23.9

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Genetic drift.
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Generation 1: Only 5 of 10 plants leave offspring (1 C super R C super R, 1 C super W C super W, and 2 C super R C super W). Generation 2: Only 2 of 10 plants leave offspring (2 C super R C super R). Generation 3: All 10 C super R C super R.

Generation 1 has 10 plants including 5 C superscript R C superscript R, 1 C superscript W C superscript W, and 4 C superscript R C superscript W. Only 5 of 10 plants leave offspring (1 C superscript R C superscript R, 1 C superscript W C superscript W, 2 C superscript R C superscript W).(p (frequency of C superscript R) = 0.7, q (frequency of C superscript W) = 0.3).

Generation 2 has 10 plants including 3 C superscript R C superscript R, 3 C superscript W C superscript W, and 4 C superscript R C superscript W. Only 2 of 10 plants leave offspring (2 C superscript R C superscript R) (p = 0.5, q = 0.5).

Generation 3 has 10 plants all of which are C superscript R C superscript R (p = 1.0, q = 0.0).

This small wildflower population has a stable size of ten plants. Suppose that by chance only five plants of generation 1 (those highlighted in yellow) produce fertile offspring. Again by chance, only two plants of generation 2 leave fertile offspring.

VISUAL SKILLS*Based on this diagram, summarize how the frequency of the*CW*allele changes over time.*

*For suggested answer, see*Appendix A.*

Certain circumstances can result in genetic drift having a significant impact on a population. Two examples are the founder effect and the bottleneck effect.

The Founder Effect

When a few individuals become isolated from a larger population, this smaller group may establish a new population whose gene pool differs from the source population; this is called the **founder effect**. The founder effect might occur, for example, when a few members of a population are blown by a storm to a new island. Genetic drift, in which chance events alter allele frequencies, can occur in such a case because the storm indiscriminately transports some individuals (and their alleles), but not others, from the source population.

The founder effect probably accounts for the relatively high frequency of certain inherited disorders among isolated human populations. For example, in 1814, 15 British colonists founded a settlement on Tristan da Cunha, a group of small islands in the Atlantic Ocean midway between Africa and South America. Apparently, one of the colonists carried a recessive allele for retinitis pigmentosa, a progressive form of blindness that affects homozygous individuals. Of the founding colonists' 240 descendants on the island in the late 1960s, four had retinitis pigmentosa. The frequency of the allele that causes this disease is ten times higher on Tristan da Cunha than in the populations from which the founders came.

The Bottleneck Effect

A sudden change in the environment, such as a fire or flood, may drastically reduce the size of a population. A severe drop in population size can cause the **bottleneck effect**, so named because the population has passed through a "bottleneck" that greatly reduces its size **(Figure 23.10)**. By chance alone, certain alleles may be overrepresented among the survivors, others may be underrepresented, and some may be absent altogether. Ongoing genetic drift is likely to have substantial effects on the gene pool until the population becomes large enough that chance events have less impact. But even if a population that has passed through a bottleneck ultimately recovers in size, it may have low levels of genetic variation for a long period of time—a legacy of the genetic drift that occurred when the population was small.

Figure 23.10

The bottleneck effect.

Original population: Bottle full of white, blue, and gold marbles. Bottlenecking effect: few marbles are released from the bottle when it is turned over. Surviving population: 5 blue, 1 white.

Shaking a few marbles through the neck of a bottle is analogous to a drastic reduction in the size of a population. By chance, blue marbles are overrepresented in the surviving population and gold marbles are absent.

Case Study: *Impact of Genetic Drift on the Greater Prairie Chicken*

Human actions sometimes create severe bottlenecks. Millions of greater prairie chickens (*Tympanuchus cupido*) once lived on the prairies of Illinois. As these prairies were converted to farmland during the 1800s and 1900s, the number of greater prairie chickens plummeted **(Figure 23.11a)**. By 1993 fewer than 50 birds remained. These few surviving birds had low levels of genetic variation, and less than 50% of their eggs hatched, compared with much higher hatching rates of the larger populations in Kansas and Nebraska **(Figure 23.11b)**.

Figure 23.11

Genetic drift and loss of genetic variation.

Genetic drift and loss of genetic variation.

Part a. Grasslands in which the greater prairie chickens live once covered most of the state of Illinois in 1820 (pre-bottleneck). In 1993, with less than 1% of the grasslands remaining, the prairie chickens were found in just two locations in Southern Illinois (post-bottleneck). The Illinois population of greater prairie chickens dropped from millions of birds in the 1800s to fewer than 50 birds in 1993.

Part b. In the small Illinois population of greater prairie chickens, genetic drift led to decreases in the number of alleles per locus and the percentage of eggs hatched.

The table is as follows: See the table: "Tables" tab

These data suggest that genetic drift during the bottleneck may have led to a loss of genetic variation and an increase in the frequency of harmful alleles. To investigate this hypothesis, researchers extracted DNA from 15 museum specimens of Illinois greater prairie chickens. Of the 15 birds, 10 had been collected in the 1930s, when there were 25,000 greater prairie chickens in Illinois, and 5 had been collected in the 1960s, when there were 1,000 greater prairie chickens in Illinois. By studying the DNA of these specimens, the researchers were able to obtain a minimum, baseline estimate of how much genetic variation was present in the Illinois population *before* the population shrank to extremely low numbers. This baseline estimate is a key piece of information that is not usually available in cases of population bottlenecks.

The researchers surveyed six loci and found that the 1993 population had fewer alleles per locus than the pre-bottleneck Illinois or the current Kansas and Nebraska populations (see Figure 23.11b). Thus, as predicted, drift had reduced the genetic variation of the small 1993 population. Drift may also have increased the frequency of harmful alleles, leading to the low egg-hatching rate. To counteract these negative effects, 271 birds from neighboring states were added to the Illinois population over four years. This strategy succeeded: New alleles entered the population, and the egg-hatching rate improved to over 90%. Overall, studies on the Illinois greater prairie chicken illustrate the powerful effects of genetic drift in small populations and provide hope that in at least some populations, these effects can be reversed.

Effects of Genetic Drift: A Summary

The examples we've described highlight four key points:

- **Genetic drift is significant in small populations.** Chance events can cause an allele to be disproportionately over- or underrepresented in the next generation. Although chance events occur in populations of all sizes, they tend to alter allele frequencies substantially only in small populations.
- **Genetic drift can cause allele frequencies to change at random.** Because of genetic drift, an allele may increase in frequency one year, then decrease the next; the change from year to year is not predictable. Thus, unlike natural selection, which in a given environment consistently favors some alleles over others, genetic drift causes allele frequencies to change at random over time.
- **Genetic drift can lead to a loss of genetic variation within populations.** By causing allele frequencies to fluctuate randomly over time, genetic drift can eliminate alleles from a population. Because evolution depends on genetic variation, such losses can influence how effectively a population can adapt to a change in the environment.
- **Genetic drift can cause harmful alleles to become fixed.** Alleles that are neither harmful nor beneficial can be lost or become fixed (reach a frequency of 100%) by chance through genetic drift. In very small populations, genetic drift can also cause alleles that are slightly harmful to become fixed. When this occurs, the population's survival can be threatened (as in greater prairie chickens).

Gene Flow

Natural selection and genetic drift are not the only phenomena affecting allele frequencies. Allele frequencies can also change by **gene flow**, the transfer of alleles into or out of a population due to the movement of fertile individuals or their gametes. For example, suppose that near our original hypothetical wildflower population there is another population consisting primarily of white-flowered individuals (CWCW). Insects carrying pollen from these plants may fly to and pollinate plants in our original population. The introduced CW alleles would modify our original population's allele frequencies in the next generation. Because alleles are transferred between populations, gene flow tends to reduce the genetic differences between populations. In fact, if it is extensive enough, gene flow can result in two populations combining into a single population with a common gene pool.

Alleles transferred by gene flow can also affect how well populations are adapted to local environmental conditions. For instance, mainland and island populations of the Lake Erie water snake (*Nerodia sipedon*) differ in their color patterns **(Figure 23.12)**: Nearly all snakes from the Ohio or Ontario mainlands are strongly banded, whereas the majority of snakes from islands are unbanded or intermediate. Banding coloration is an inherited trait, determined by a few loci (with alleles that encode bands being dominant to alleles that encode the absence of bands). On islands, water snakes live along rocky shorelines, while on the mainland, they live in marshes. Snakes without bands are more well camouflaged in island habitats than are snakes with bands. Hence, on islands, snakes without bands survive at higher rates than do snakes with bands.

Figure 23.12

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    Gene flow and local adaptation in the Lake Erie water snake (*Nerodia
sipedon*).
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Bar chart compares banding patterns of Lake Erie water snake populations in the Ontario mainland, Ohio mainland, and islands in Lake Erie.

A map shows there are several small islands in Lake Erie between the mainland of Ohio and Ontario. A bar chart compares the banding patterns of Lake Erie water snake populations (N. sipedon) in the Ontario mainland, Ohio mainland, and islands in Lake Erie.

The x-axis shows banding patterns A, B, C, and D for the Ohio mainland, the islands, and the anterior mainland. The y-axis shows percentage of individuals, ranging from 0 to 100. For the Ohio mainland population, the percentages are approximately 83% (banding pattern A), 11% (banding pattern B), 2% (banding pattern C), and 4% (banding pattern D). For the islands, the percentages are approximately 5% (pattern A), 16% (pattern B), 30% (pattern C), 49% (pattern D). For the Ontario Maitland, 100% of individuals are banding pattern A.

Researchers assigned letters to variations in coloration in *N. sipedon* populations. Color pattern A is strong banding, patterns B and C are intermediate banding, and pattern D is no banding. Banding is advantageous for camouflage in mainland environments, whereas having no bands is advantageous in island environments. However, gene flow from the mainland causes banding to persist in island populations.

WHAT IF?**Suppose a severe weather event caused island populations to decrease in size but did not affect the size of mainland populations. Predict how gene flow from the mainland would affect color patterns in island populations in the year immediately after the storm. Explain.*

*For suggested answer, see**Appendix A*.*

These data indicate that snakes without bands are favored by natural selection in island populations. Thus, we might expect that *all* snakes on islands would lack bands. Why is this not the case? The answer lies in gene flow from the mainland. In any given year, three to ten snakes from the mainland swim to the islands and join the populations there. As a result, each year such migrants transfer alleles for banded coloration from the mainland (where nearly all snakes have bands) to the islands. This ongoing gene flow has prevented selection from removing all of the alleles for banded coloration from island populations —thereby preventing island populations from adapting fully to local conditions.

Gene flow can also transfer alleles that improve the ability of populations to adapt to local conditions. For example, gene flow has resulted in the worldwide spread of several insecticide resistance alleles in the mosquito *Culex pipiens*, a vector of West Nile virus and other diseases. Each of these alleles has a unique genetic signature that allowed researchers to document that it arose by mutation in only one or a few geographic locations. In their population of origin, these alleles increased because they provided insecticide resistance. These alleles were then transferred by gene flow to other populations exposed to insecticides, where again, their frequencies increased as a result of natural selection.

Finally, gene flow has become an increasingly important agent of evolutionary change in human populations. Humans today move much more freely about the world than in the past. As a result, mating is more common between members of populations that previously had very little contact, leading to an exchange of alleles and fewer genetic differences between those populations.

BioFlix® Animation

Natural Selection, Genetic Drift, and Gene Flow

image

Play

image

Play

Concept Check 23.3

- In what sense is natural selection more “predictable” than genetic drift?
- Distinguish genetic drift from gene flow in terms of (a) how they occur and (b) their implications for future genetic variation in a population.
- **WHAT IF?** Suppose two plant populations exchange pollen and seeds. In one population, individuals of genotype AA are most common (9,000AA,900Aa,100aa), while the opposite is true in the other population (100AA,900 Aa,9,000aa). If neither allele has a selective advantage, what will happen over time to the allele and genotype frequencies of these populations?

*For suggested answers, see*Appendix A.**

Concept 23.4: Natural selection is the only mechanism that consistently causes adaptive evolution

Evolution by natural selection is a blend of chance and “sorting”: chance in the creation of new genetic variations (as in mutation) and sorting as natural selection favors some alleles over others. Because of this favoring process, the outcome of natural selection is *not* random. Instead, natural selection consistently increases the frequencies of alleles that provide reproductive advantage, thus leading to adaptive evolution.

Natural Selection: *A Closer Look*

To see how natural selection can cause adaptive evolution, we’ll begin with the concept of relative fitness and the different ways that selection acts on an organism’s phenotype.

Relative Fitness

The phrases “struggle for existence” and “survival of the fittest” are commonly used to describe natural selection, but these expressions are misleading if always taken to mean direct competitive contests among individuals. There *are* animal species in which individuals, usually the males, lock horns or otherwise do combat to determine mating privilege. But reproductive success is generally more subtle and depends on many factors besides outright battle. For example, a barnacle that is more efficient at collecting food than its neighbors may have greater stores of energy and hence be able to produce a larger number of eggs. A moth may have more offspring than other moths in the same population because its body colors more effectively conceal it from predators, improving its chance of surviving long enough to produce more offspring. These examples illustrate how in a given environment, certain traits can lead to greater **relative fitness**: the contribution an individual makes to the gene pool of the next generation *relative to* the contributions of other individuals.

Although we often refer to the relative fitness of a genotype, remember that the entity that is subjected to natural selection is the whole organism, not the underlying genotype. Thus, selection acts more directly on the phenotype than on the genotype; it acts on the genotype indirectly, via how the genotype affects the phenotype.

Directional, Disruptive, and Stabilizing Selection

Natural selection can alter the frequency distribution of heritable traits in three ways, depending on which phenotypes in a population are favored: through directional selection, disruptive selection, and stabilizing selection.

Directional selection occurs when conditions favor individuals exhibiting one extreme of a phenotypic range, thereby shifting a population's frequency curve for the phenotypic character in one direction or the other **(Figure 23.13a)**. Directional selection is common when a population's environment changes or when members of a population migrate to a new (and different) habitat. For instance, an increase in the relative abundance of large seeds over small seeds led to an increase in beak depth in a population of Galápagos finches (see Figure 23.2).

Figure 23.13

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Modes of selection.
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Frequency of individuals versus Phenotypes (fur color) for the Original population. The x-axis is Phenotypes and the y-axis is Frequency of individuals. The curve is a bell shaped population curve.

These cases describe three ways in which a hypothetical deer mouse population with heritable variation in fur coloration might evolve. The graphs show how the frequencies of individuals with different fur colors change over time. The large white arrows symbolize selective pressures against certain phenotypes.

Frequency of individuals versus Phenotypes (fur color) for the Original population. The x-axis is Phenotypes and the y-axis is Frequency of individuals. The curve is a bell shaped curve with white mice at the left tail of the bell and dark brown mice at the right tail of the bell. Shades of intermediate color mice are at the center of the curve with the highest frequency of individuals.

MAKE CONNECTIONS*Review*Figure 22.13*. Which mode of selection has occurred in soapberry bug populations that feed on the introduced golden rain tree? Explain.*

*For suggested answer, see*Appendix A.**

****Disruptive selection(Figure 23.13b)** occurs when conditions favor individuals at both extremes of a phenotypic range over individuals with intermediate phenotypes. One example is a population of black-bellied seedcracker finches in Cameroon whose members display two distinctly different beak sizes. Small-billed birds feed mainly on soft seeds, whereas large-billed birds specialize in cracking hard seeds. It appears that birds with intermediate-sized bills are relatively inefficient at cracking both types of seeds and thus have lower relative fitness.

Stabilizing selection(Figure 23.13c) acts against both extreme phenotypes and favors intermediate variants. This mode of selection reduces variation and tends to maintain the status quo for a particular phenotypic character. For example, the birth weights of most human babies lie in the range of 3–4 kg (6.6–8.8 pounds). Babies who are much smaller suffer considerably higher rates of mortality; babies who are much larger show a slight increase in mortality.

Regardless of which type of selection occurs, however, the basic mechanism remains the same. Selection favors individuals whose heritable phenotypic traits provide higher reproductive success than do the traits of other individuals.

HHMI BioInteractive Video

Selection for Tuskless Elephants

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The Key Role of Natural Selection in Adaptive Evolution

The adaptations of organisms include many striking examples. Certain octopuses, for example, have the ability to change color rapidly, enabling them to blend into different backgrounds. Another example is the remarkable jaws of snakes **(Figure 23.14)**, which allow them to swallow prey much larger than their own head (a feat analogous to a person swallowing a whole watermelon). Other adaptations, such as a version of an enzyme that shows improved function in cold environments, may be less visually dramatic but just as important for survival and reproduction.

Figure 23.14

Movable jaw bones in snakes.

The bones of the upper jaw of a snake are movable via ligament at the back of the jaw.

The bones of the upper jaw of a snake are movable via ligament at the back of the jaw. The skull bones of most terrestrial vertebrates are relatively rigidly attached to one another, limiting jaw movement. In contrast, most snakes have movable bones in their upper jaw, allowing them to swallow food much larger than their head. Photo shows a snake swallowing an egg larger than its head.

Such adaptations can arise gradually over time as natural selection increases the frequencies of alleles that enhance survival or reproduction. As the proportion of individuals that have favorable traits increases, the degree to which a species is well suited for life in its environment improves; that is, adaptive evolution occurs. However, the physical and biological components of an organism's environment may change over time. As a result, what constitutes a "good match" between an organism and its environment can be a moving target, making adaptive evolution a continuous, dynamic process. Environmental conditions can also differ from place to place, causing different alleles to be favored in different locations. When this occurs, natural selection can cause the populations of a species to differ genetically from one another.

And what about genetic drift and gene flow? Both can, in fact, increase the frequencies of alleles that enhance survival or reproduction, but neither does so consistently. Genetic drift can cause the frequency of a slightly beneficial allele to increase, or, alternately, to decrease. Similarly, gene flow may introduce alleles that are advantageous or ones that are disadvantageous. Natural selection is the only evolutionary mechanism that consistently leads to adaptive evolution.

HHMI BioInteractive Video

Got Lactase? The Co-evolution of Genes and Culture

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Sexual Selection

Charles Darwin was the first to explore the implications of **sexual selection**, a process in which individuals with certain inherited characteristics are more likely than other individuals of the same sex to obtain mates. Sexual selection can result in **sexual dimorphism**, a difference in secondary sexual characteristics between males and females of the same species **(Figure 23.15)**. These distinctions include differences in size, color, ornamentation, and behavior.

Figure 23.15

Sexual dimorphism and sexual selection.

Peacock with erect tail that forms a shimmering fan displayed to the peahen.

A peacock (left) and a peahen (right) show extreme sexual dimorphism. There is intrasexual selection between competing males, followed by intersexual selection when the females choose among the showiest males.

How does sexual selection operate? There are several ways. In **intrasexual selection**, meaning selection within the same sex, individuals of one sex compete directly for mates of the opposite sex. In many species, intrasexual selection occurs among males. For example, a single male may patrol a group of females and prevent other males from mating with them. The patrolling male may defend his status by defeating smaller, weaker, or less fierce males in combat. More often, this male is the psychological victor in ritualized displays that discourage would-be competitors but do not risk injury that would reduce his own fitness (see Figure 51.16). Intrasexual selection also occurs among females in a variety of species, including ring-tailed lemurs and broadnosed pipefish.

In **intersexual selection**, also called *mate choice*, individuals of one sex (usually the females) are choosy in selecting their mates from the other sex. In many cases, the female's choice depends on the showiness of the male's appearance or behavior (see Figure 23.15). What intrigued Darwin about mate choice is that male showiness may not seem adaptive in any other way and may in fact pose some risk. For example, bright plumage may make male birds more visible to predators. But if such characteristics help a male gain a mate, and if this benefit outweighs the risk from predation, then both the bright plumage and the female preference for it will be reinforced because they enhance overall reproductive success.

How do female preferences for certain male characteristics evolve in the first place? One hypothesis is that females prefer male traits that are correlated with "good genes." If the trait preferred by females is indicative of a male's overall genetic quality, both the male trait and female preference for it should increase in frequency. **Figure 23.16** describes one experiment testing this hypothesis in gray tree frogs.

Figure 23.16

Inquiry: Do females select mates based on traits indicative of "good genes"?

Experiment

Female gray tree frogs (*Hyla versicolor*) prefer to mate with males that give long mating calls. To test whether the genetic makeup of long-calling (LC) males is superior to that of short-calling (SC) males, researchers fertilized half the eggs of each female with sperm from an LC male and the other half with sperm from an SC male. In two separate experiments (one in 1995, one in 1996), the resulting half-sibling offspring were raised in a common environment, and their survival and growth were monitored.

A diagram of an experiment. The eggs of a female gray tree frog are fertilized with sperm from a short call male and long call male. Survival and growth of the half-sibling offspring from both fathers were then compared.

Results

Do females select mates based on traits indicative of “good genes”?

Offspring performance	1995	1996
larval survival	long call better	no significant difference
larval growth	no significant difference	long call better
time to metamorphosis	long call better, shorter	long call better, shorter

NSD=no significant difference; LC better=offspring of LC males superior to offspring of SC males.

Conclusion

Because offspring fathered by an LC male outperformed their half-siblings fathered by an SC male, the team concluded that the duration of a male’s mating call is indicative of the male’s overall genetic quality. This result supports the hypothesis that female mate choice can be based on a trait that indicates whether the male has “good genes.”

Data from A. M. Welch et al., Call duration as an indicator of genetic quality in male gray tree frogs, *Science* 280:1928–1930 (1998).

INQUIRY IN ACTION Read and analyze the original paper in *Inquiry in Action: Interpreting Scientific Papers*.

WHAT IF?*Why did the researchers split each female frog’s eggs into two batches for fertilization by different males? Why didn’t they mate each female with a single male frog?*

*For suggested answer, see*Appendix A*.*

In several bird species, the traits preferred by females are related to overall male health. Here, too, female preference appears to be based on traits that reflect “good genes”—in this case, alleles indicative of a robust immune system.

Balancing Selection

As we’ve seen, genetic variation is often found at loci affected by selection. What prevents natural selection from reducing the variation at those loci by culling all unfavorable alleles? As mentioned earlier, in diploid organisms, many unfavorable recessive alleles persist because they are hidden from selection when in heterozygous individuals. In addition, selection itself may preserve variation at some loci, thus maintaining two or more phenotypic forms in a population. Known as **balancing selection**, this type of selection includes frequency-dependent selection and heterozygote advantage.

Frequency-Dependent Selection

In **frequency-dependent selection**, the fitness of a phenotype depends on how common it is in the population. Consider the scale-eating fish (*Perissodus microlepis*) of Lake Tanganyika, in Africa. These fish attack other fish from behind, darting in

to remove a few scales from the flank of their prey. Of interest here is a peculiar feature of the scale-eating fish: Some are “left-mouthed” and some are “right-mouthed.”

Because their mouth twists to the left, left-mouthed fish typically attack their prey’s right flank **(Figure 23.17)**. (To see why, twist your lower jaw and lips to the left and imagine trying to take a bite from the left side of a fish, approaching it from behind.) Similarly, right-mouthed fish attack from the left. Prey species guard against attack from whatever phenotype of scale-eating fish is most common. Thus, from year to year, selection favors whichever mouth phenotype is least common. As a result, the frequency of left- and right-mouthed fish oscillates over time, and this balancing selection keeps the frequency of each phenotype close to 50%.

Figure 23.17

Frequency-dependent selection.

Graph of Frequency of left-mouthed individuals versus sample year. Data are: 1981, 0.55; 1983, 0.45; 1985, 0.6; 1987, 0.5; 1988, 0.45; 1989, 0.5; 1990, 0.55.

In a population of the scale-eating fish Perissodus microlepis, the frequency of left-mouthed individuals (red data points) rises and falls in a regular manner. The black data points indicate the frequency of left-mouthed individuals among adults that reproduced in three sample years.

Graph of Frequency of left-mouthed individuals versus sample year. Sample year is on the x-axis from 1981 to 1990. Frequency of left-mouthed individuals from 0 to 1.0 is on the y-axis. Data is: 1981, 0.55; 1983, 0.45; 1985, 0.6; 1987, 0.5; 1988, 0.45; 1989, 0.5; 1990, 0.55. Data for frequency of individuals that reproduce is 1981, 0.45; 1987, 0.6; 1990, 0.35.

INTERPRET THE DATA*For 1981, 1987, and 1990, compare the frequency of left-mouthed individuals among breeding adults to the frequency of left-mouthed individuals in the entire population. What do the data indicate about when natural selection favors left-mouthed individuals over right-mouthed individuals (or vice versa)? Explain.*

Heterozygote Advantage

If individuals who are heterozygous at a particular locus have greater fitness than do both kinds of homozygotes, they exhibit **heterozygote advantage**. In such a case, natural selection tends to maintain two or more alleles at that locus. Note that heterozygote advantage is defined in terms of *genotype*, not phenotype. Thus, whether heterozygote advantage represents stabilizing or directional selection depends on the relationship between the genotype and the phenotype. For example, if the phenotype of a heterozygote is intermediate to the phenotypes of both homozygotes, then heterozygote advantage is a form of stabilizing selection.

An example of heterozygote advantage occurs at the locus in humans that codes for the β polypeptide subunit of hemoglobin, the oxygen-carrying protein of red blood cells. In homozygous individuals, a recessive allele at that locus causes sickle-cell disease. The red blood cells of people with sickle-cell disease become distorted in shape, or *sickled*, under low-oxygen conditions (see Figure 5.19). These sickled cells can clump together and block the flow of blood in the capillaries, damaging organs such as the kidney, heart, and brain. Although some red blood cells become sickled in heterozygotes, not enough become sickled to cause sickle-cell disease.

As described in **Figure 23.18**, heterozygotes for the sickle-cell allele are protected against the most severe effects of malaria, a disease caused by a parasite that infects red blood cells. One reason for this partial protection is that the body destroys sickled red blood cells rapidly, killing the parasites they harbor. Malaria is a major killer in some tropical regions. In such regions,

selection favors heterozygotes over homozygous dominant individuals, who are more vulnerable to the effects of malaria, and also over homozygous recessive individuals, who develop sickle-cell disease. These selective pressures have caused the frequency of the sickle-cell allele to reach relatively high levels in areas where the malaria parasite, which is carried by mosquitoes, is common.

Figure 23.18 Interactive Visual Activity

Make Connections: The Sickle-Cell Allele.

Interact with this figure in the Figure 23.18 Interactive Visual Activity: Make Connections: The Sickle-Cell Allele.

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Figure 23.18

AP Make Connections: The Sickle-Cell Allele

An illustration depicting the impact od a harmful allele in sickle-cell disease.

A two-part illustration is shown.

In the top half of the image, a child with sickle-cell disease is shown. Sickle-cell disease is a genetic disorder that strikes individuals who have two copies of the sickle-cell allele. This allele causes an abnormality in the structure and function of hemoglobin, the oxygen-carrying protein in red blood cells. Although sickle-cell disease is lethal if not treated, in some regions the sickle-cell allele can reach frequencies as high as 15 to 20 percent. How can such a harmful allele be so common?

Events at the molecular level:

- Due to a point mutation, the sickle-cell allele differs from the wild-type allele by a single nucleotide. (See Figure 17.26)
- The resulting change in one amino acid leads to hydrophobic interactions between the sickle-cell hemoglobin proteins under low-oxygen conditions.
- As a result, the sickle-cell proteins bind to each other in chains that together form a fiber.

A chromosome with the sickle-cell allele unwinds into the DNA sequence. A base substitution in the template strand, replacing thymine with adenine causes a codon change during transcription. This, in turn, changes the amino acid sequence of hemoglobin from glutamic acid to valine, causing a change in the shape of hemoglobin into a fiber-like structure. The consequence of this is a sickle-shaped red blood cell.

In the bottom half of the image, a chromosome with the wild-type allele unwinds into the DNA sequence. The template is normal and causes the formation of normal globular hemoglobin. The consequence of this is a normal red blood cell.

Consequences for cells: The abnormal hemoglobin fibers distort the red blood cell into a sickle shape under low-oxygen conditions, such as those found in small blood vessels returning to the heart.

An illustration depicting the effects of the sickle-cell allele on individual organisms and populations is depicted.

Effects on Individual Organisms

- The formation of sickled red blood cells causes homo-zygotes with two copies of the sickle-cell allele to have sickle-cell disease.
- Some sickling also occurs in heterozygotes, but not enough to cause the disease; they have sickle-cell trait. (See Figure 14.17)

A blood vessel with sickled blood cells of a homozygote blocking small blood vessels is shown. This can cause great pain and damage to organs such as the heart, kidney, and brain.

A blood vessel with normal red blood cells, which are flexible and able to flow freely through small blood vessels, is shown.

Evolution in Populations

- Homozygotes with two sickle-cell alleles are strongly selected against because of mortality caused by sickle-cell disease. In contrast, heterozygotes experience few harmful effects from sickling yet are more likely to survive malaria than are homozygotes.
- In regions where malaria is common, the net effect of these opposing selective forces is heterozygote advantage. This has caused evolutionary change in populations—the products of which are the areas of relatively high frequencies of the sickle-cell allele shown in the map below:

A view of the world mainly showing Africa, Europe, and Asia with different levels of frequencies of the sickle-cell allele is represented. Few, small regions of central Africa show greater than 15.0% frequency, western Africa shows 9.0% - 12.0% frequency, parts of northern and central Africa, Saudi Arabia, and central and southern India show 3.0% - 9.0% frequency of the sickle cell allele. Distribution of malaria caused by Plasmodium falciparum (a parasitic, unicellular eukaryote, carried by mosquitoes) is found all over Africa and India and the western parts of Saudi Arabia.

MAKE CONNECTIONS*In a region free of malaria, would individuals who are heterozygous for the sickle-cell allele be selected for or selected against? Explain.*

*For suggested answer, see*Appendix A.**

HHMI BioInteractive Video

The Making of the Fittest: Natural Selection in Humans (Sickle-Cell Disease)

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Why Natural Selection Cannot Fashion Perfect Organisms

Though natural selection leads to adaptation, nature abounds with examples of organisms that are less than ideally suited for their lifestyles. There are several reasons why.

- **Selection can act only on existing variations.** Natural selection favors only the fittest phenotypes among those currently in the population, which may not be the ideal traits. New advantageous alleles do not arise on demand. For example, historically, snowshoe hares (*Lepus americanus*) have molted, changing their coats from brown to white, at a time of year that matched the onset of snowfall, providing camouflage all winter. But due to climate change, the first snowfall now occurs later in the year. In some populations, the date at which the hares change their coat color has remained the same, causing the hares to be poorly camouflaged early in winter and thus easier for predators to spot

and kill **(Figure 23.19)**. Because their gene pools lack alleles that could delay when molting occurs, these populations have been unable to adapt to changing conditions.

Figure 23.19

Lack of variation in a population can limit adaptation.

Part a. A brown hare blends in with its environment. Part b. A white hair stands out in its brown environment.

*For the snowshoe hare, changing its coat color from brown **(a)** to white **(b)** too early is disadvantageous, but the population lacks alleles that would encode a delay in molting.*

- **Evolution is limited by historical constraints.** Each species has a legacy of descent with modification from ancestral forms. Evolution does not scrap the ancestral anatomy and build each new complex structure from scratch; rather, evolution co-opts existing structures and adapts them to new situations. We could imagine that if a terrestrial animal were to adapt to an environment in which flight would be advantageous, it might be best just to grow an extra pair of limbs that would serve as wings. However, evolution does not work this way; instead, it operates on the traits an organism already has. Thus, in birds and bats, an existing pair of limbs took on new functions for flight as these organisms evolved from nonflying ancestors.
- **Adaptations are often compromises.** Each organism must do many different things. A seal spends part of its time on rocks; it could probably walk better if it had legs instead of flippers, but then it would not swim nearly as well. We humans owe much of our versatility and athleticism to our prehensile hands and flexible limbs, but these also make us prone to sprains, torn ligaments, and dislocations: Structural reinforcement has been compromised for agility. Organisms face many such *trade-offs* in which the ability to perform one function may reduce the ability to perform another—and as with seals and humans, those trade-offs can restrict adaptive evolution.
- **Chance, natural selection, and the environment interact.** Chance events can affect the subsequent evolutionary history of populations. For instance, when a storm blows insects or birds hundreds of kilometers over an ocean to an island, the wind does not necessarily transport those individuals that are best suited to the new environment. Thus, not all alleles present in the founding population's gene pool are better suited to the new environment than the alleles that are "left behind." In addition, the environment at a particular location may change unpredictably from year to year, again limiting the extent to which adaptive evolution results in organisms being well suited for current environmental conditions.

With these four constraints, evolution does not result in perfect organisms. Natural selection operates on a "better than" basis. We can, in fact, see evidence for evolution in the many imperfections of the organisms it produces.

Concept Check 23.4

- What is the relative fitness of a sterile mule? Explain.
- Explain why natural selection is the only mechanism that consistently leads to adaptive evolution in a population.
- **VISUAL SKILLS** Consider a population in which heterozygotes at a certain locus have an extreme phenotype (such as being larger than homozygotes) that confers a selective advantage. Compare this description to the three types of selection shown in Figure 23.13. Does this situation represent directional, disruptive, or stabilizing selection? Explain.

*For suggested answers, see*Appendix A.**